



A novel simulation technique for UWB-FMCW radar: enhancing through-the-wall detection capabilities

Ahmed M. Eid¹ · Mouaaz Nahas²

Received: 17 February 2025 / Accepted: 26 September 2025
© The Author(s) 2025

Abstract

Ultrawideband frequency-modulated continuous-wave (UWB-FMCW) radar is a solution for through-the-wall (TTW) imaging and detection technology. TTW radar is an attractive UWB microwave imaging system that comprises radars capable of localizing and monitoring humans in hazardous locations or under unfavorable conditions. This study presents a novel approach for simulating the RF front-end transceiver of a UWB-FMCW radar system. The proposed methodology is implemented on a 9 GHz FMCW radar, utilizing a single antenna for both transmission and reception. The simulation model proposed in the paper is shown to be a valuable tool for designing UWB-FMCW through-wall imaging radar systems.

Keywords Ultrawideband · Frequency modulated · Continuous wave · Through-the-wall · Radar

1 Introduction

Radar is a detection system that uses radio waves to measure the distance, height, direction, or velocity of objects. In other words, a radar system is a stand-alone device employed to determine the position and velocity of a static or moving object by examining its characteristics (Alotaibi 2022). One of the well-known radar systems is the frequency-modulated continuous-wave (FMCW) radar, which is typically used for imaging purposes (Schellenberg et al. 2007; Yi'an et al. 2007; Asensio-López et al. 2004; Cooper et al. 2008). For example, automotive FMCW radar is utilized to measure the speed of vehicles that violate traffic regulations on roads (Winkler 2007; Ali and Vossiek 2010). It can also serve as a driver assistance system to improve driving conditions and prevent accidents, and also for ship navigation/identification (Steinhauer et al. 2008; Park and Kim 2011). Moreover, FMCW radars are capable of detecting targets even in environments with significant ground clutter (Duarte et al. 2007). Other applications of FMCW include measuring wind speed and direction in geosciences (Nekrasov and Hooeboom

2005), planet's surface analysis (Leuschen et al. 2002), monitoring airplanes and missiles in military environments (Jahagirdar 2012), maritime navigation (Huang et al. 2011), and monitoring wind turbine blades (Moll et al. 2018).

Ultrawideband (UWB) see-through-wall (STW) imaging is a compelling application of UWB technology (Muqaibel et al. 2005; Yang et al. 2008). The STW imaging operates by transmitting and receiving UWB pulses in the time domain through the use of synthetic aperture radar (SAR) (Yang and Fathy 2007). Through the acquisition of reasonably high-resolution images, this sort of imaging may be used in the quick detection of persons behind walls as well as rescue operations in avalanches and fallen structures. The STW radar can easily detect objects through highly dense materials including concrete blocks, fiberglass, wood, and bricks (see (Yang and Fathy 2007) for more details).

The UWB signals utilize extremely narrow monocycle Gaussian pulses, typically lasting only a few nanoseconds (Ghavami et al. 2007). Because these pulses are extremely short in time, they provide an extensive bandwidth spectrum in the frequency domain. Moreover, given that UWB signals consist of electromagnetic pulses of sub-nanosecond durations, they can easily penetrate biological tissues and provide high spatial resolution (Tan and Chia 2004). UWB sensing has also been used in biomedical applications to monitor patients remotely (Le et al. 2020). In all such applications, the signaling is required to be robust, efficient, secure, and of high performance.

✉ Mouaaz Nahas
mmnahas@uqu.edu.sa

¹ Suez Canal Authority, Suez 41515, Ismailia, Egypt

² Department of Electrical Engineering, Umm Al-Qura University, 21955 Makkah, Saudi Arabia

Through-the-wall (TTW) radar technology is used to detect, count, locate, and track individuals within buildings by using frequencies in the range of UHF to S-band (Maaref et al. 2009). When studying TTW radars, the influence of walls on radar signals must be predicted, and the TTW radar waves must be able to penetrate different kinds of walls (e.g., brick, concrete block, rough concrete, and plasterboard) in order to sense the presence of a person or target behind the wall (Maaref and Millot 2012). A simulation methodology was developed for an RF front-end transceiver of an FMCW radar system to measure both the range and velocity of a specific target (Eid and Nahhas 2013). The methodology was implemented on an FMCW radar operating at 2.45 GHz (i.e., operating at the S-band). On the contrary, the frequency band considered in our study is the X-band (in particular, the 9 GHz frequency). At this frequency, various studies considered the development of UWB-FMCW radars for detecting stationary or moving objects behind walls (for example, (Hussein et al. 2022; Hu et al. 2019; Acar et al. 2021)). The “multi-input-multi-output” (MIMO) technology (Ahmad et al. 2022) was adopted in many researches in this area (Huang et al. 2011; Hu et al. 2019) and other areas such as sensor networks (Alotaibi 2019) and RFID (Alotaibi et al. 2022), due to its numerous advantages. More recent studies have made significant contributions to the topics addressed in this paper. For instance, the study in Wang et al. (2024) explores advancements in algorithm design aimed at enhancing target identification from multipath ghosts and clutter in SISO IR-UWB radar systems. Additionally, new techniques for interference suppression in FMCW radar systems produced promising outcomes, especially with the implementation of feature-oriented unsupervised networks (Zhang et al. 2023). Furthermore, improvements in radar technologies have greatly advanced through-the-wall human activity recognition, offering valuable insights for various applications, including security and surveillance (Yousaf et al. 2024). Also, the development of innovative W-band through-wall radar systems, particularly those employing sector scanning and traveling wave tubes, demonstrated considerable potential for improving signal penetration capabilities (Tian et al. 2024). Scientists at the Pacific Northwest National Laboratory (PNNL) have developed advanced millimeter-wave scanners for security checkpoints like airports. These systems employ a technique called microwave holography, capturing detailed phase and amplitude data across a surface to produce sharp images of concealed weapons and prohibited items (Sheen et al. 2001). In the research detailed in Aftanas et al. (2008), ultra-wideband (UWB) maximum length binary sequence (M-sequence) signals were used to see through a brick wall and image objects. The researchers successfully scanned and created images of three different types of objects hidden behind the wall. Their method included a straightforward approach to account for the fact

that radio waves travel at different speeds inside the wall. By measuring how thick the wall was and its permittivity, they could precisely pinpoint and image the objects in their actual locations (Aftanas et al. 2008).

The work presented in this paper proposes a novel simulation methodology for an RF front-end transceiver of a UWB-FMCW TTW imaging radar system using Advanced Design System (ADS) software. The main novelty of this research lies in its simulation approach for the RF front-end of a UWB-FMCW system. This simulation uniquely incorporates a human-wall model that considers how the electrical properties of both the wall and human tissues change with frequency. While the paper provides a foundational analysis for creating a UWB microwave sensor for detecting objects behind walls, it does not delve into the specifics of the FMCW radar front-end design, radar back-end processing, antenna system, or control mechanisms.

2 Basic theoretical concepts of UWB-FMCW radars

Various technologies can be employed to construct a UWB TTW radar (Maaref et al. 2009). FMCW radar offers several benefits for TTW applications, with the primary advantage being its superior power efficiency and signal-to-noise ratio.

The operation of the FMCW radar is based on continuous sweeping from an initial frequency to a final frequency (*Chirp Signal*). Figure 1 shows the time domain for the FMCW radar signal.

Figure 2 shows the time-domain amplitude plot of the transmitted signal during a typical FMCW sweep. It is assumed that the frequency of the transmitted signal changes while its amplitude stays constant. This makes the plot of the

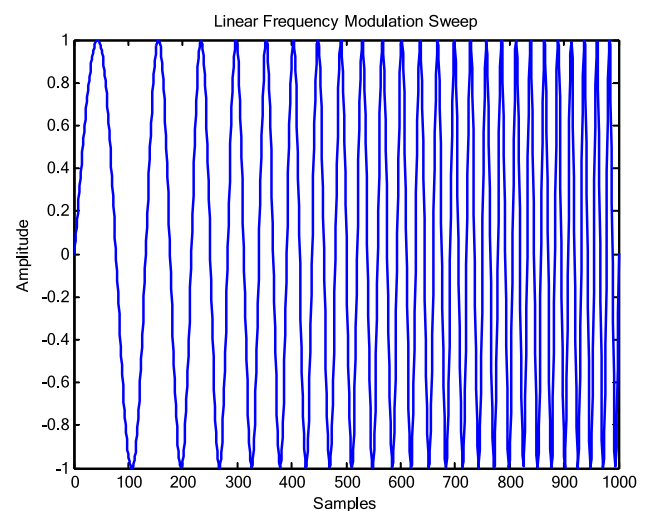


Fig. 1 Time domain signal for the FMCW radar

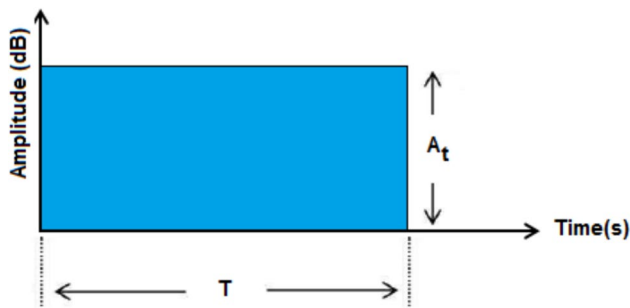


Fig. 2 Plot of the transmitted signal time-domain amplitude during an FMCW sweep

amplitude seem like a frequency-modulated pulse waveform, as illustrated in the figure. If the received signal is undistorted, its shape would match the transmitted signal but with a different amplitude, as depicted in Fig. 3.

Figure 4 illustrates the block diagram of a UWB-FMCW through-the-wall radar system. It mainly consists of a transmitter and a receiver besides a target model. The transmitter path begins with a linear FM chirp waveform generator. The generator output passes to a power splitter

that feeds the mixer of the receiving path. The other output from the splitter goes to a circulator before being directed to the antenna. The signal that is transmitted by the antenna hits the target, which is in this case a person standing behind the wall.

The reflected signal is shaped by the target (delayed, attenuated, and frequency shifted), and then sent back to the antenna. The signal received by the antenna goes through the circulator and is then directed to a low noise amplifier (IF Amplifier) before reaching the mixer. After that, the signal is multiplied by the original transmitted signal. The mixer will then output and go through a low-pass filter (IF filter) to suppress the harmonic frequencies produced by the mixer.

Figure 5 shows the range determination of a stationary target. As shown in the figure, the first ramp (from the left) represents the transmitted signal produced by the signal generator, whereas the second ramp (from the left) represents the received (i.e., reflected) signal from the target.

Let us assume that a reflecting object exists which is located at a distance $r=r_o$, the frequency modulated (FM) carrier signal can be expressed as follows:

$$u_s = A_s \cos \phi_s(t) \quad (1)$$

Fig. 3 Plot of the received signal time-domain amplitude during an FMCW sweep

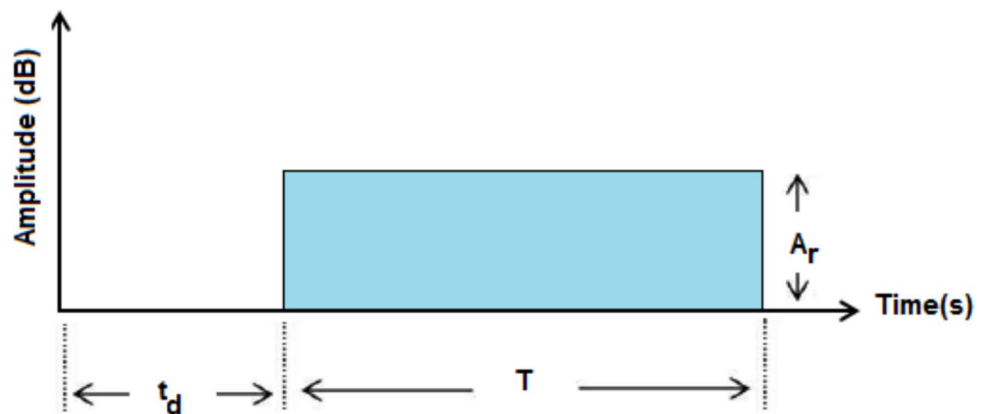


Fig. 4 FMCW radar block diagram

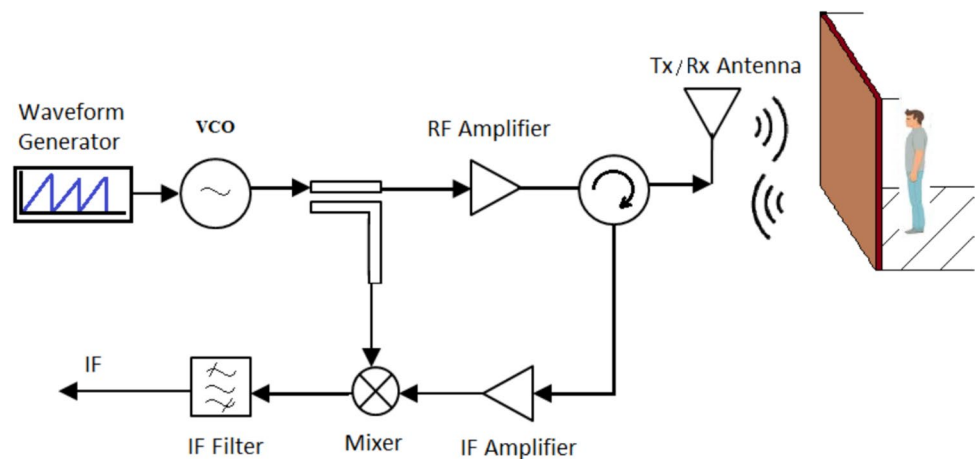
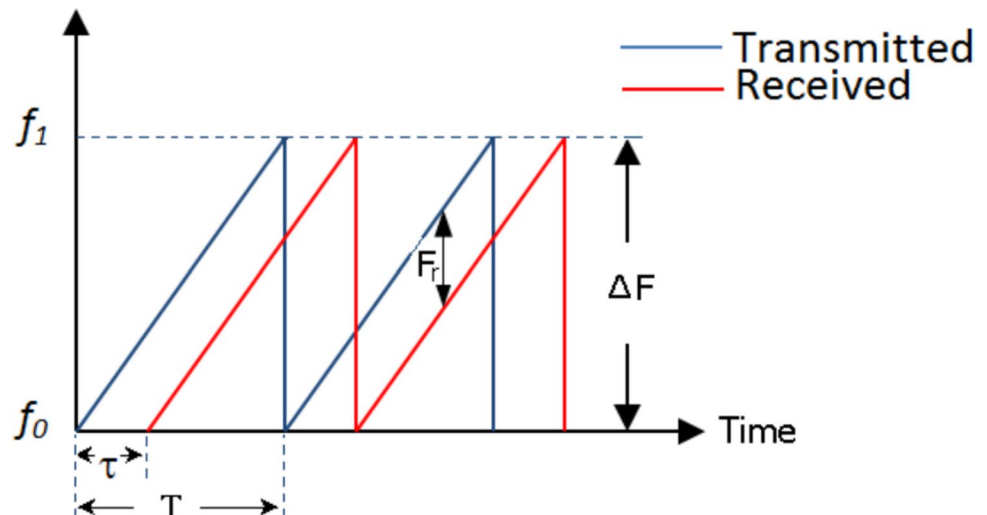


Fig. 5 Range determination of a stationary target



$$f_s(t) = f_o + \frac{\Delta F}{T}t \text{ for } 0 \leq t \leq T \quad (2)$$

$$\Phi_s(t) = 2\pi \int_0^t f_s(t)dt + \varphi_{so} = 2\pi \left(f_o t + \frac{1}{2} \cdot \frac{\Delta F}{T} t^2 \right) + \varphi_{so} \quad (3)$$

where A_s is the signal amplitude, f_o is the carrier frequency, φ_{so} is the initial phase, and $f_s(t)$ is a time-varying frequency. After the transit time τ , an echo signal is returned. We can express the phase of the received signal as follows:

$$\varphi_E(t) = \varphi_s(t - \tau) = 2\pi \left(f_o(t - \tau) + \frac{1}{2} \cdot \frac{\Delta F}{T}(t - \tau)^2 \right) - \varphi_{so} \quad (4)$$

A beat note is formed when the echo signal is heterodyned (i.e., mixed) with a part of the transmitter signal in a nonlinear device (e.g., a diode). The beat note component can be expressed as follows (Han 2014; Skolnik 1970):

$$u_d(t) = U_D \cos(\varphi_s(t) - \varphi_E(t)) \quad (5)$$

$$u_d(t) = U_D \cos \left[2\pi \left[\frac{\Delta F}{T} t \tau + f_o \tau - \frac{1}{2} \Delta F \tau \frac{\tau}{T} \right] \right] \quad (6)$$

where U_D is the beat note signal amplitude. If there is no shift in Doppler frequency (i.e., nonmoving target), the beat note (i.e., the difference frequency) measures the target range. The transit time τ and the beat note frequency F_r are calculated as follows:

$$\tau = \frac{2r_o}{c_o} \quad (7)$$

$$F_r = \frac{\Delta F}{T} \tau \quad (8)$$

where c_o is the velocity of light. The measured beat note frequency resolution limit ΔF_r is determined using the time span T as follows:

$$\Delta F_r = \frac{1}{T} \quad (9)$$

If two targets with ranges r_1 and r_2 exist, their beat note frequencies are estimated by Eqs. (7) and (8) as follows:

$$F_{r1} = \frac{\Delta F}{T} \cdot \frac{2r_1}{c_o} \quad (10)$$

$$F_{r2} = \frac{\Delta F}{T} \cdot \frac{2r_2}{c_o} \quad (11)$$

Therefore, the beat note frequency resolution limit can be obtained by the following equation:

$$\Delta F_r = F_{r2} - F_{r1} = \frac{\Delta F}{T} \cdot \left(\frac{2r_2}{c_o} - \frac{2r_1}{c_o} \right) \quad (12)$$

By comparing Eqs. (9) and (12), the measured range resolution limit can be found using the formula:

$$\Delta r = r_2 - r_1 = \frac{c_o}{2\Delta F} \quad (13)$$

3 Selection of frequency band

Based on the relationships described in Section 2 (particularly Eqs. (8) and (13)), the resolution of the radar is directly influenced by the signal bandwidth and the speed of propagation in the medium. Selecting the appropriate frequency band is crucial for system design. Microwaves are often favored over millimeter waves due to their lower spatial sampling

requirements, which result in reduced costs for the antenna system, quicker data collection, and less computational power needed for processing. The optimal frequency band is determined by the resolution required for the specific application. Radar resolution, which measures the ability to distinguish between closely spaced targets, is a key factor. For detecting humans through walls, the aim is to identify the presence and general shape of a person without revealing detailed anatomical features. Millimeter-wave resolution provides high detail but may raise privacy concerns, while centimeter-wave resolution offers a good balance by providing sufficient information while maintaining ethical standards.

In wideband radar systems, the resolution of the produced 3D image is affected by factors such as operational bandwidth, center frequency, target distance, and the size of the antenna array's aperture. Importantly, the resolution in the down-range direction is exclusively determined by the radar's operational bandwidth. In line with Eq. (13), which defines range resolution in terms of bandwidth, the down-range resolution δ_{dr} can be expressed as

$$\delta_{dr} = \frac{C}{2B} \quad (14)$$

where C represents the speed at which electromagnetic waves propagate through the sensing medium, while B refers to the operational bandwidth measured at the -10 dB point. The equation demonstrates that achieving finer resolution in the down-range direction necessitates a larger bandwidth. This underscores the significance of ultra-wideband (UWB) technology for applications demanding high down-range resolution. For example, a bandwidth of at least 6 GHz is required to attain a resolution of 2.5 cm in this direction. On the other hand, the cross-range resolution, typically considered in terms of height and width, is calculated using a different formula that follows:

$$\delta_{el} = \lambda \frac{R}{L_{el}} \quad (15)$$

$$\delta_{az} = \lambda \frac{R}{L_{az}} \quad (16)$$

where L_{el} and L_{az} denote the lengths of the effective aperture of the antenna array in the elevation and azimuth directions, respectively. R indicates the distance to the target, and λ is the wavelength corresponding to the central frequency. For a MIMO array, the effective aperture length L is calculated as the combined length of the transmit array L_{tx} and the receive array L_{Rx} .

$$L = L_{tx} + L_{Rx}$$

Assuming the transmit and receive antenna apertures are of equal dimensions, the cross-range resolutions can be expressed as follows:

$$\delta_{el} = \lambda \frac{R}{L_{elTx} + L_{elRx}} = \frac{\lambda}{2} \frac{R}{L_{el}} \quad (17)$$

$$\delta_{az} = \lambda \frac{R}{L_{azTx} + L_{azRx}} = \frac{\lambda}{2} \frac{R}{L_{az}} \quad (18)$$

As an example, using Eq. (18), to obtain centimeter-level resolution δ_{az} in the azimuth direction at a range of $R = 1$ m within a 3D image, an antenna aperture of at least $L_{az} = 1$ m is required. This level of resolution also necessitates a central wavelength of approximately $\lambda = 2$ cm, which corresponds to a frequency of around 15 GHz. Based on these preliminary calculations, an operational frequency range of 6 to 18 GHz is selected.

4 Methodology

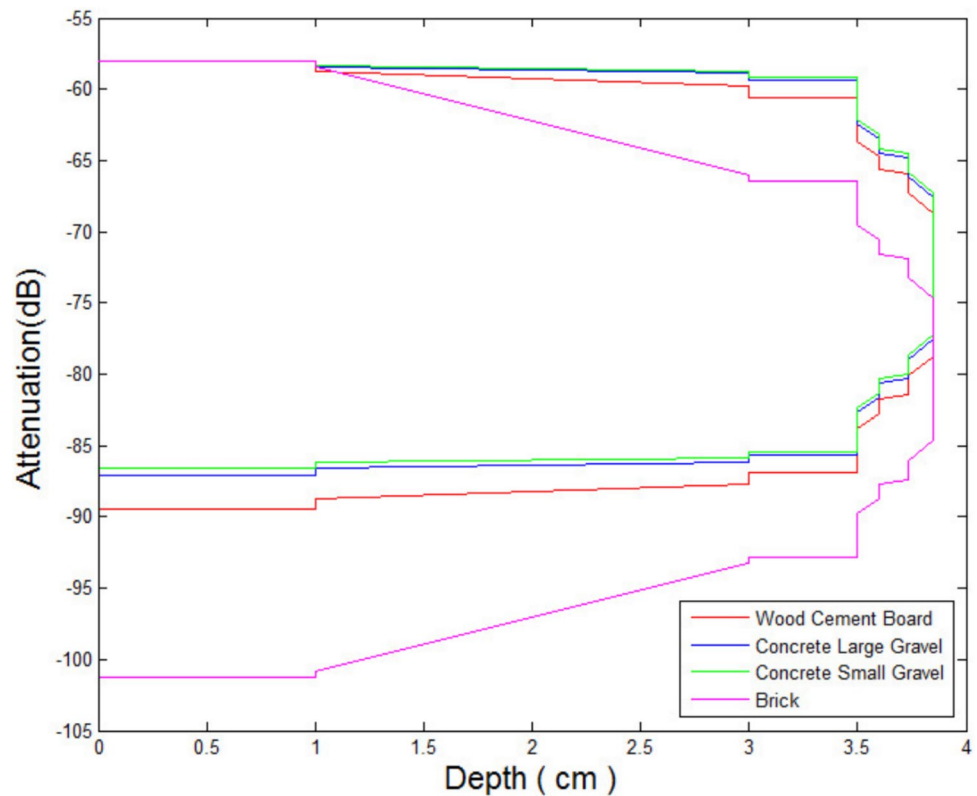
This section provides the practical method used in this study, which includes both the model proposed for the electromagnetic 1 d human-wall and the linear FM chirp waveform generation. The initial step in our system is to evaluate its capability to detect weaker objects amidst a strong reflection, which is expected to primarily originate from the human body.

4.1 Electromagnetic 1 d human-wall model

The weakening of signals due to wave propagation through wall materials greatly impacts the performance of through-the-wall radar systems. Therefore, a thorough understanding of how walls affect radar signals is essential for analyzing these systems. To achieve this, we propose using detailed electromagnetic (EM) wall models to evaluate their interaction with radar waves. Our research specifically examines wave propagation through walls made of brick, wood cement board, small concrete, and large concrete.

Initially, a basic one-dimensional multilayer electromagnetic model is employed. Wall materials are dispersive, meaning their dielectric properties change with frequency. The model uses the relative permittivity and conductivity of common building materials such as brick ($\epsilon_r = 3.75$, $\sigma = 0.03$), wood cement board ($\epsilon_r = 5.7$, $\sigma = 0.075$), small concrete ($\epsilon_r = 3.4$, $\sigma = 0.02$), and large concrete ($\epsilon_r = 4$, $\sigma = 0.025$) (Zhekov et al. 2020) to accurately simulate wave propagation and attenuation. Figure 6 illustrates the attenuation of electromagnetic pulses as they travel from the radar antenna, through the wall, and back. As the figure shows, the

Fig. 6 Round-trip loss of electromagnetic pulse for various wall materials



signal's propagation is influenced by the material properties it encounters. The complex relative permittivity of construction materials significantly affects how electromagnetic (EM) waves propagate through them. Since each dielectric material has unique, frequency-dependent properties, this dispersion can be particularly important for ultra-wideband (UWB) systems. Dispersion may cause signal distortion and severely impact system performance. Figure 6 depicts the round-trip signal loss for various building materials.

As an example, a simulation scenario is considered where a test subject (a person, in this case) stands ahead of the radar antenna at a distance that is contained between 1 and 5 cm. The wall is considered the brick wall with a relative permittivity of $\epsilon_r = 3.75$.

The human body behind the wall is modeled as a three-layer stack (i.e., skin-fat-muscles) that is infinite in the direction transverse to the propagation direction, and hence, the wave propagates in a direction perpendicular to the borders

between the three layers. The electrical properties of tissues at different frequencies are given in Table 1.

Tissue dielectric properties vary considerably with frequency, making them highly dispersive. Table 1 presents the electrical characteristics of skin, fat, and muscle at frequencies of 3, 6, and 10 GHz. These values will be utilized in the subsequent models.

The electromagnetic pulse radiated from the UWB-FMCW radar's transmitting antenna will partly be reflected at every interface. The attenuation suffered by a putative electromagnetic pulse along its way to the person behind the wall and back out is plotted in Fig. 7.

This model demonstrates a round-trip loss of 100 dB for the signal to hit the human target and then return to the receiver. The transmission of radar signals through walls and signal fading in the propagation path cause significant attenuation, necessitating sufficient energy for a practical dynamic range. FMCW radar offers a superior power budget, enhanced by pulse integration but reduced by a low-duty cycle. The chirp signal achieves the highest signal-to-noise

Table 1 Electrical properties of body tissues at various frequencies (Gabriel et al. 1996)

	3 GHz		6 GHz		10 GHz	
	ϵ_r	σ (s/m)	ϵ_r	σ (s/m)	ϵ_r	σ (s/m)
Skin	37.45	1.74	34.9	3.89	31.3	8.01
Fat	5.2239	0.13	4.94	0.31	4.6	0.59
Muscles	52.05	2.14	48.2	5.20	42.8	10.6

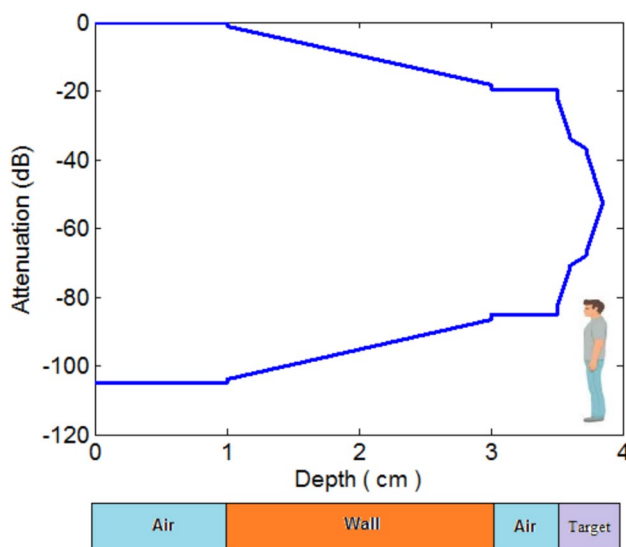


Fig. 7 Round-trip loss of electromagnetic pulse for the human target behind a wall

ratio. Importantly, by adjusting the frequency sweep duration, we would be able to optimize the received electromagnetic energy to match the target's stationary period.

The signal-to-noise ratio (SNR), a vital measure of radar performance, indicates the level of noise contamination in a signal. The necessary dynamic range is typically determined by the difference in strength between the strongest received signal and the weakest target reflection. In frequency-modulated continuous-wave (FMCW) through-the-wall (TTW) radar, a significant challenge is that the strongest signal often comes from the initial wall reflection, which is much stronger than the backscatter from a human target.

Achieving dynamic ranges above 100 dB is challenging and requires careful design of amplification, filtering, and signal digitalization stages. The radar detector must handle this wide range to prevent the strong reflection from the initial wall from overwhelming the receiver. Another approach to mitigate wall reflections is to subtract two data acquisitions taken in quick succession. For optimal performance, fast and repeatable oscillators are essential. Voltage-controlled oscillators (VCOs) and yttrium iron garnet (YIG) frequency synthesizers, which are now affordable and high-quality, will be evaluated in the next phase of research.

Figure 8 shows the block diagram of an FMCW radar system model when realized using the ADS simulation software.

ADS is a versatile software suite designed for simulating circuits and RF systems to meet specific design goals. It supports various simulation types, including DC, AC, S-parameter, Harmonic Balance, Circuit Envelope, Transient, and RF-Budget analysis. In our project, we specifically used the Circuit Envelope simulation feature. The Circuit Envelope simulator is specialized for modeling high-frequency

components like amplifiers, mixers, and oscillators, as well as subsystems that process modulated or transient RF signals. It excels in analyzing circuits with digitally modulated signals by focusing on transient analysis around the carrier frequency and its harmonics. By integrating time-domain and frequency-domain analysis methods, the Circuit Envelope simulation provides fast and accurate results for complex signals like digitally modulated RF signals. It represents input waveforms in the frequency domain using RF carriers while capturing their modulation “envelopes” in the time domain.

In ADS simulation, the attenuation is commonly modeled using an attenuator, whereas travel time—and consequently, the target's distance—can be modeled using a time-delay component.

4.2 Linear FM chirp waveform generation

Our system utilizes the fundamental Frequency-Modulated Continuous-Wave (FMCW) technique, employing a single antenna for both transmitting and receiving signals. The waveform generator is crucial for radar performance. Typically, an FMCW signal is generated by a Voltage-Controlled Oscillator (VCO), with the output frequency governed by the input voltage. Ideally, this relationship should be linear, but real-world VCOs often show deviations that mainly degrade down-range resolution. A non-linear frequency sweep results in a varying beat frequency even for stationary targets. The Microlambda MLMH-0208 is an example of a commercially available VCO known for its linear frequency sweep.

The RF pulse power source is utilized in the system-level simulations to create a pulse-modulated RF carrier with an optional frequency chirping. The RF pulse power source produces a 3–9 GHz chirp signal with 10 dBm output power. The frequency domain of the RF signal has a pulse period equal to 5000 ns, a start frequency of the sweep equal to 6 GHz, and a chirp frequency bandwidth of 6 GHz, as shown in Fig. 9.

To have the desired UWB spectrum, a frequency multiplier is placed after the RF pulse power source to provide the 6–18 GHz desired sweep. The frequency multiplier receives an input signal and generates an output spectrum containing designated spectral harmonics of that signal, as shown in Fig. 10.

5 Results and discussion

In this section, we showcase and discuss the simulation results of the UWB-FMCW system model developed in this study.

5.1 UWB-FMCW system model and simulation

In Fig. 8 above, the RF parameters of the modules are obtained from L3 Harris Narda-MITEQ's commercial

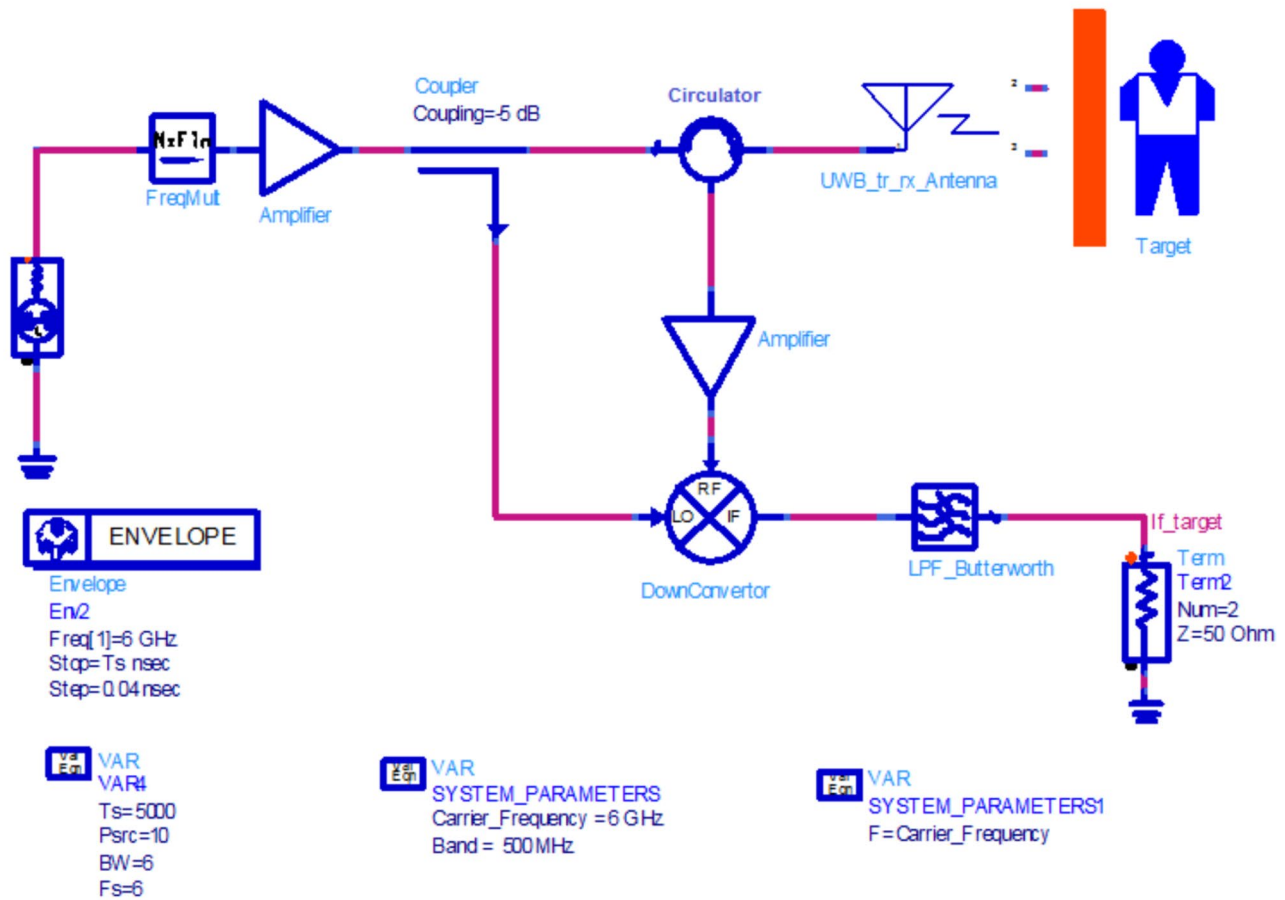


Fig. 8 Block diagram of an FMCW radar system model realized on ADS

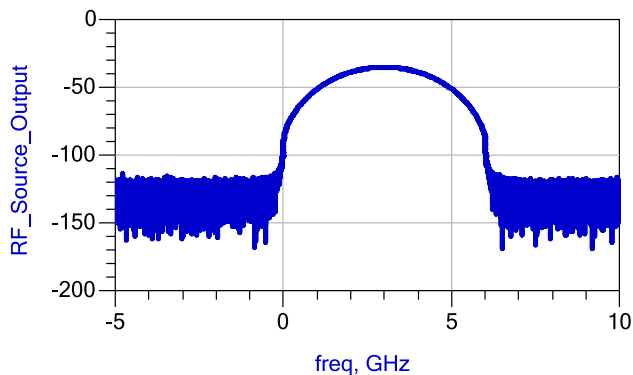


Fig. 9 Frequency spectrum of the produced RF pulse

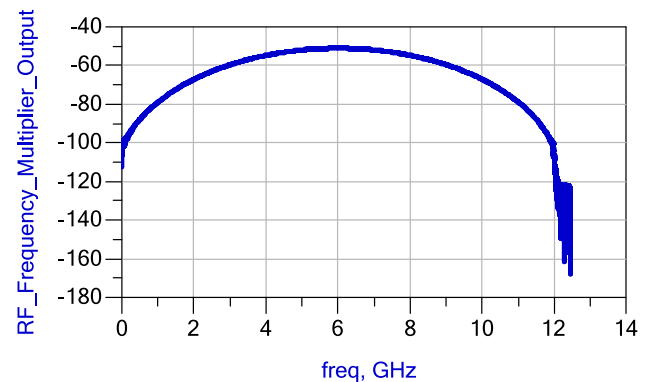


Fig. 10 The RF pulse spectrum with input spectral harmonics

off-the-shelf components. An example of these modules is the frequency multiplier model (MAX2M030180) with the following characteristics: input frequency ranges from 1.5 to 9 GHz, output frequency ranges from 3 to 18, and input power ranges from 8 to 12 dBm. Another module is

the power amplifier model (AMF-8B -06001800-60-31p), which has the operating frequency from 6 to 18 GHz, the gain of 31 dB, the output power at 1db compression of 31, and the VSWR input/output of 2:2. On the other hand, the directional coupler model (4203-10) has been used, which

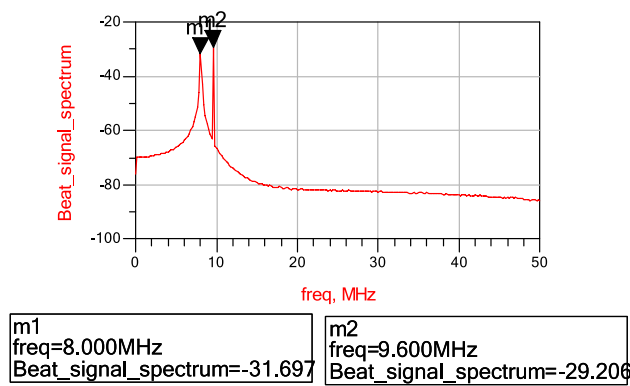


Fig. 13 The spectrum of the beat signal for two targets

at 0.6 m. Using Eq. (11), the peaks in the beat signal due to the first and second targets are calculated as follows:

$$F_{beat1} = \frac{12 \times 10^9}{5 \times 10^{-6}} \cdot \frac{2 \times 0.5}{3 \times 10^8} = 8MHz$$

$$F_{beat2} = \frac{12 \times 10^9}{5 \times 10^{-6}} \cdot \frac{2 \times 0.6}{3 \times 10^8} = 9.6MHz$$

The simulation results align with the theoretical analysis as shown in Fig. 13, where a peak in the beat signal is observed at 8 MHz for the target at 0.5 m from the wall and 9.6 MHz for the target at 0.6 m from the wall.

6 Conclusion

This study explores potential solutions for detecting humans behind walls using ultrawideband frequency-modulated continuous-wave (UWB-FMCW) radars. A complete RF front-end transceiver of the radar system model has been developed and evaluated using ADS simulation software. It was successfully applied to a 9 GHz FMCW radar. Nonetheless, other circuit simulators like SPICE can also be used to simulate the proposed model. The radar system model consists of typical transmitting and receiving paths as well as a target model (that has been developed and fully described). Results showed that the simulation model was an effective tool for designing UWB-FMCW radar systems, highlighting its potential use for localizing and monitoring humans in hazardous locations or under adverse conditions. It is worth noting that the system model is adaptable to various realistic effects, including different channel models. The channel model can be placed between the antenna and the target model to account for the attenuation and noise introduced by the channel. This research can lead to faster and more efficient development of UWB-FMCW through-wall imaging

radar systems, and future studies can focus on improving the methodology and exploring other potential applications. A more in-depth characterization of the chosen RF components would result in more accurate outcomes. Additionally, broadening the study to encompass the detectability of a wider array of targets could enhance the findings. In this research, we concentrated on detecting a person behind a brick wall, but the spectrum of potential threats is much broader, including scenarios from an unarmed individual to one carrying a weapon.

Acknowledgements The authors extend their appreciation to Umm Al-Qura University, Saudi Arabia, for funding this research work through grant number: 25UQU4300346GSSR01.

Author contribution Ahmed M. Eid: conceptualization, methodology, writing—original draft, supervision. Mouaaz Nahas: data curation, formal analysis, writing—review and editing, visualization.

Funding This research work was funded by Umm Al-Qura University, Saudi Arabia under grant number: 25UQU4300346GSSR01.

Data availability The datasets generated and/or analysed during the current study are included in this published article.

Declarations

Ethical approval and consent to participate Not applicable.

Consent for publication Not applicable.

Competing interests The authors declare no competing interests.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

- Acar YE, Saritas I, Yaldiz E (2021) An S-band zero-IF SFCW through-the-wall radar for range, respiration rate, and DOA estimation. *Measurement* 186:110221. <https://doi.org/10.1016/j.measurement.2021.110221>
- Aftanas M, Zaikov E, Drutarovsky M, Sachs J (2008) Through wall imaging of the objects scanned by M-sequence UWB radar system. In: 2008 18th International Conference Radioelektronika. IEEE, Prague, pp 1–4

- Ahmad F, Bin Abbas W, Khalid S et al (2022) Performance enhancement of mmWave MIMO systems using machine learning. *IEEE Access* 10:73068–73078. <https://doi.org/10.1109/ACCESS.2022.3190388>
- Ali F, Vossiek M (2010) Detection of weak moving targets based on 2-D range-Doppler FMCW radar Fourier processing. In: *German Microwave Conference Digest of Papers*. pp 214–217
- Alotaibi M (2019) PAPR reduction scheme using OFDM multicarrier signals smart constellation in wireless sensor networks. *Umm Al-Qura Univ J Eng Archit* 9:31–43
- Alotaibi M (2022) Multi-criteria decision making to detect multiple moving targets in radar using digital codes. *Sensors (Basel)* 22:3176. <https://doi.org/10.3390/s22093176>
- Alotaibi M, Murad M, Alhuthali SAH et al (2022) MIMO radio frequency identification: a brief survey. *Sensors* 22:4115. <https://doi.org/10.3390/s22114115>
- Asensio-López A, Blanco-del-Campo A, Gismero-Menoyo J et al (2004) High range-resolution radar scheme for imaging with tunable distance limits. *Electron Lett* 40:1085–1086. <https://doi.org/10.1049/el:20045552>
- Cooper KB, Dengler RJ, Chattopadhyay G et al (2008) A high-resolution imaging radar at 580 GHz. *IEEE Microw Wirel Compon Lett* 18:64–66. <https://doi.org/10.1109/LMWC.2007.912049>
- Duarte CC, Dorta Naranjo BP, Lopez AA, Del Campo AB (2007) Cwlfm radar for ship detection and identification. *IEEE Aerosp Electron Syst Mag* 22:22–26. <https://doi.org/10.1109/MAES.2007.323295>
- Eid AM, Nahhas AM (2013) System simulation of RF front-end transceiver for frequency modulated continuous wave radar. *Int J Comput Appl* 75:16–23. <https://doi.org/10.5120/13082-0332>
- Gabriel S, Lau RW, Gabriel C (1996) The dielectric properties of biological tissues: III. Parametric models for the dielectric spectrum of tissues. *Phys Med Biol* 41:2271–2293. <https://doi.org/10.1088/0031-9155/41/11/003>
- Ghavami M, Michael L, Kohno R (2007) *Ultra wideband signals and systems in communication engineering*. John Wiley & Sons
- Han JH (2014) Study on 24 GHz short range radar system using delayed FMCW signal. Kwangwoon University
- Hu Z, Zeng Z, Wang K et al (2019) Design and analysis of a UWB MIMO radar system with miniaturized Vivaldi antenna for through-wall imaging. *Remote Sens* 11:1867. <https://doi.org/10.3390/rs11161867>
- Huang Y, Brennan PV, Patrick D et al (2011) FMCW based MIMO imaging radar for maritime navigation. *PIER* 115:327–342. <https://doi.org/10.2528/PIER11021509>
- Hussein K, Zekry A, Abouelatta M, et al (2022) Low-cost L-band FMCW radar for motion detection behind multiple walls. In: *International Conference on Computing, Control and Industrial Engineering*. Springer, pp 694–702
- Jahagirdar DR (2012) A high dynamic range miniature DDS-based FMCW radar. In: *2012 IEEE Radar Conference*. pp 0870–0873
- Le M, Le D-K, Lee J (2020) Multivariate singular spectral analysis for heartbeat extraction in remote sensing of UWB impulse radar. *Sens Actuators, A* 306:111968. <https://doi.org/10.1016/j.sna.2020.111968>
- Leuschen C, Kanagaratnam P, Yoshikawa K, et al (2002) Field experiments of a surface-penetrating radar for Mars. In: *IEEE International Geoscience and Remote Sensing Symposium*. pp 3579–3581
- Maaref N, Millot P, Pichot C, Picon O (2009) A study of UWB FM-CW radar for the detection of human beings in motion inside a building. *IEEE Trans Geosci Remote Sens* 47:1297–1300. <https://doi.org/10.1109/TGRS.2008.2010709>
- Maaref N, Millot P (2012) Array-based UWB FMCW through-the-wall radar. In: *Proceedings of the 2012 IEEE International Symposium on Antennas and Propagation*. pp 1–2
- Moll J, Simon J, Malzer M et al (2018) Radar imaging system for in-service wind turbine blades inspections: initial results from a field installation at a 2 MW wind turbine. *Prog Electromagn Res* 162:51–60. <https://doi.org/10.2528/PIER18021905>
- Muqaibel A, Safaai-Jazi A, Bayram A et al (2005) Ultrawideband through-the-wall propagation. *IEE Proc-Microwaves, Antennas Propag* 152:581–588
- Nekrasov A, Hoogeboom P (2005) A ka-band backscatter model function and an algorithm for measurement of the wind vector over the sea surface. *IEEE Geosci Remote Sens Lett* 2:23–27. <https://doi.org/10.1109/LGRS.2004.840613>
- Park S-G, Kim Y-H (2011) Desing of a low resolution FMCW radar for small target detection under ground clutter. In: *2011 3rd International Asia-Pacific Conference on Synthetic Aperture Radar (APSAR)*. pp 1–1
- Schellenberg J, Chedester R, McCoy J (2007) Multi-channel receiver for an E-band FMCW imaging radar. In: *2007 IEEE/MTT-S International Microwave Symposium*. pp 1359–1362
- Sheen DM, McMakin DL, Hall TE (2001) Three-dimensional millimeter-wave imaging for concealed weapon detection. *IEEE Trans Microw Theory Tech* 49:1581–1592. <https://doi.org/10.1109/22.942570>
- Skolnik MI (1970) *Radar handbook*. McGraw-Hill Book Co., New York
- Steinhauer M, Ruoss H-O, Irion H, Menzel W (2008) Millimeter-wave-radar sensor based on a transceiver array for automotive applications. *IEEE Trans Microw Theory Tech* 56:261–269. <https://doi.org/10.1109/TMTT.2007.914635>
- Tan AEC, Chia MYW (2004) UWB radar transceiver and measurement for medical imaging. In: *IEEE International Workshop on Biomedical Circuits and Systems*, 2004. p S3/1–9
- Tian Z, Zhang J, Yi S (2024) Innovative W-band through-wall radar with sector scanning: utilizing traveling wave tubes for enhanced penetration. *IEEE Sens J* 24:30801–30809. <https://doi.org/10.1109/jsen.2024.3438276>
- Wang S, Wang H, Wang G et al (2024) Algorithm design to identify targets from multipath ghosts and clutters for SISO IR-UWB radar: incorporating waveform features and trajectory. *IEEE Sens J* 24:2278–2288. <https://doi.org/10.1109/jsen.2023.3336679>
- Winkler V (2007) Range Doppler detection for automotive FMCW radars. In: *2007 European Radar Conference*. pp 166–169
- Yang Y, Zhang C, Fathy AE (2008) Development and implementation of ultra-wideband see-through-wall imaging system based on sampling oscilloscope. *IEEE Antennas Wireless Propag Lett* 7:465–468. <https://doi.org/10.1109/LAWP.2008.2000829>
- Yang Y, Fathy A (2007) Design and implementation of a low-cost real-time ultra-wide band see-through-wall imaging radar system. In: *2007 IEEE/MTT-S International Microwave Symposium*. pp 1467–1470
- Yi'an Z, Zhiping Y, Weidong C, Dongjin W (2007) Two-dimensional radar imaging based on instant range profile. In: *2007 1st Asian and Pacific Conference on Synthetic Aperture Radar*. pp 236–239
- Yousaf J, Yakoub S, Karkanawi S et al (2024) Through-the-wall human activity recognition using radar technologies: a review. *IEEE Open J Antennas Propag* 5:1815–1837. <https://doi.org/10.1109/ojap.2024.3459045>
- Zhang H, Wei S, Wang M et al (2023) Fuas-net: feature-oriented unsupervised network for FMCW radar interference suppression. *IEEE Trans Microw Theory Tech* 72:2602–2619. <https://doi.org/10.1109/tmtt.2023.3318669>
- Zhekov SS, Franek O, Pedersen GF (2020) Dielectric properties of common building materials for ultrawideband propagation studies [measurements corner]. *IEEE Antennas Propag Mag* 62:72–81. <https://doi.org/10.1109/MAP.2019.2955680>